# Nicholas D. Gray, Ph.D.

**DATA SCIENTIST**

ndgray00@gmail.com | 513-504-0179 | Maryland Heights, MO | github.com/ndg13b

Data scientist with nearly four years of industry experience in insurance analytics, built on a decade of quantitative research training. Combines a Ph.D. in Cognitive Psychology with hands-on expertise in SQL, Python, and machine learning to turn large, messy data sources into models, reports, and recommendations that stakeholders act on.

## TECHNICAL SKILLS

| | |
|---|---|
| **Languages & Tools** | Python (advanced), SQL / Microsoft SQL Server, R, MATLAB, Microsoft Excel (advanced) |
| **Machine Learning** | XGBoost & gradient boosting, Regression (linear & logistic), Predictive modeling, Model evaluation |
| **Data & Analysis** | ETL & data cleaning, Exploratory data analysis, Statistical inference, Experimental design, Survey & qualitative analysis |
| **Communication** | Stakeholder reporting, Data visualization, Technical writing, Teaching & presentation |

## PROFESSIONAL EXPERIENCE

### Data Science Associate
Oct 2022 – Present

*Safety National Casualty Corporation · St. Louis, MO*

- Build analytics reports and dashboards used by internal teams and external stakeholders across the business.
- Extract and join data from large, complex sources in Microsoft SQL Server, then transform and clean it in Python for analysis.
- Develop machine-learning models — including XGBoost — that produce business-relevant predictions for underwriting and claims questions.
- Design custom analyses out of exploratory investigation, tailoring method and reporting to the decision each stakeholder actually faces.

### Postdoctoral Researcher — Quantitative Analysis
Aug 2019 – Oct 2022

*Institute for Successful Longevity, Florida State University · Tallahassee, FL*

- Led statistical analysis for large-scale behavioral clinical trials on healthy cognitive aging, working across longitudinal datasets and multiple analysis platforms.
- Established statistical performance norms and built regression models identifying the variables that drive cognitive outcomes.
- Integrated qualitative streams — surveys and interviews — with quantitative measures to produce findings neither would support alone.
- Designed and programmed novel studies end to end and published the results in peer-reviewed journals.

### Graduate Researcher
Aug 2013 – May 2019

*Department of Psychology, Florida State University · Tallahassee, FL*

- Modeled experimental data with logistic regression and related statistical methods across several software platforms.
- Designed, programmed, and ran grant-funded experimental research; managed the lab, recruiting and supervising undergraduate researchers.
- Taught laboratory sections of Cognitive Psychology and co-instructed Careers in Psychology (2014–2019), translating complex statistical concepts for non-expert audiences.

## SELECTED PROJECTS

**adherence** — `Python · github.com/ndg13b/adherence`
A statistics package that scores how regular someone's engagement is — not just how much of it there is — and turns that into a forecast of when they will next show up.

**CivicGateway** — `JavaScript · civicgateway.org`
A nonpartisan site showing upcoming elections and elected officials by city; static front end over a Supabase/PostgreSQL schema with row-level security and an offline fallback.

**tuned-in** — `JavaScript · ndg13b.github.io/tuned-in`
A teaching collection pairing pop-culture clips with the cognitive psychology they demonstrate — and, just as often, get wrong.

## EDUCATION

**Ph.D., Cognitive Psychology** 2019
Florida State University
*Thesis: The causes of and influences on being reminded*

**M.S., Cognitive Psychology** 2016
Florida State University
*Thesis: Avoiding interference: Differentiation and reminding*

**B.S., Psychology, cum laude** 2012
Ohio State University

**Additional coursework:** Machine Learning with Python; Microsoft SQL Server (independently completed)

## SELECTED PUBLICATIONS

Gray, N., & Charness, N. (2022). Technology obsolescence across the adult lifespan in a USA internet sample. Frontiers in Public Health, 10, 1005822.

Gray, N., Yoon, J.-S., Charness, N., Boot, W. R., et al. (2021). Relative effectiveness of general versus specific cognitive training for aging adults. Psychology and Aging, 37(2), 210–221.

Carr, D. C., Tian, S., He, Z., et al., incl. Gray, N. (2022). Motivation to engage in aging research: Are there typologies and predictors? The Gerontologist, gnac035.

Charness, N., Boot, W. R., & Gray, N. (2020). Mobile monitoring and intervention technology (MMI) for adaptive aging. In Mobile Technology for Adaptive Aging: Proceedings of a Workshop (pp. 21–40). National Academies Press.

*Conference presentations at the Psychonomic Society (2018) and the Cognitive Aging Conference (2022).*